Reproduce-then-Extend — Fatal Militarized Disputes (Gibler & Braithwaite) (Day 2)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. Gibler, D. M. & Braithwaite, A., "Hostile Contiguous States" (Harvard Dataverse doi:10.7910/DVN/HQLOOJ) — a binary-response regression for whether a dyadic militarized dispute becomes **fatal**, on dyadic predictors. This is *interstate* conflict, distinct from the civil-war cases; fatal disputes are rare (~1.3%) → scored by **AUC-PR**.

1 Background

Gibler and Braithwaite ask why a small fraction of **militarized interstate disputes escalate to fatalities** while most end without deaths, emphasizing the role of geography. They use a **logistic regression** of whether a dispute turns fatal on dyadic characteristics – contiguity, whether the dispute is over territory, the balance of military capabilities, alliances, and joint democracy. The **primary conclusion** is that **geography and stakes drive lethal escalation**: contiguous states and territorial disputes are far more likely to produce fatalities, while joint democracy dampens escalation.

2 Data and codebook

Unit of analysis: a militarized interstate dispute-dyad; n = 20,263 (~1.3% fatal).

Variable	Definition
`fatal`	1 if the militarized dispute produced at least one battle death **(outcome)**
`lowdem`	lower of the two states' Polity democracy scores (weakest-link)
`terr500k10yr`	1 if the dispute is over territory
`defense`	1 if the two states share a defensive alliance
`caprat`	national-capability ratio (stronger to weaker state, CINC)
`contig`	1 if the states are geographically contiguous
`peaceyears`	years since the dyad's previous dispute
`spline1-spline3`	cubic-spline bases for peace-years (temporal dependence)
`interaction4`, `pnbdem_low`	constructed interaction terms from the published specification

3 Descriptive analysis

```r
d <- read.csv(paste0(
  "https://raw.githubusercontent.com/bdesmarais/intro",
  "-ml-statistical-horizons-4day/main/tutorials/day4_",
  "classic_learners/05_gibler_braithwaite/data/gb20.c",
  "sv"))
preds <- setdiff(names(d), "fatal")
preds <- preds[sapply(d[preds], function(x) length(unique(x)) > 1)]
clean <- function(v) gsub("[._]+", " ", v)   # tidy names for plots
```

3.1 Dataset and provenance

Unit of analysis. One row is a **militarized interstate dispute-dyad** — a pair of states involved in a militarized dispute. The data ship with the published replication package (Gibler & Braithwaite, Harvard Dataverse doi:10.7910/DVN/HQLOOJ) and describe the cross-national universe of interstate disputes rather than a sample, so there is no sampling frame to correct for.

```r
c(rows = nrow(d), columns = ncol(d),
  n_predictors = length(preds), n_fatal = sum(d$fatal),
  complete_rows = sum(complete.cases(d)))
```

```
##          rows       columns  n_predictors       n_fatal complete_rows
##         20263            12            11           263         20263
```

The frame has 20263 disputes and 12 columns — a **binary outcome** (`fatal`) and 11 numeric predictors. Only two predictors are indicator (0/1) variables (`contig`, `defense`); the rest are counts or continuous ratios. A variable dictionary:

Variable	Role	Type	Coding / description
`fatal`	outcome	binary	1 if the dispute produced >= 1 battle death
`contig`	predictor	binary	1 if the two states are geographically contiguous
`terr500k10yr`	predictor	count	intensity of territorial stakes in the dispute
`defense`	predictor	binary	1 if the dyad shares a defensive alliance
`caprat`	predictor	ratio	national-capability ratio (CINC), 0-1
`lowdem`	predictor	integer	weaker state's Polity democracy score (-10..10)
`pnbdem_low`	predictor	continuous	democracy-based interaction term
`interaction4`	predictor	integer	constructed interaction from the published model
`peaceyears`	predictor	count	years since the dyad's previous dispute
`spline1-spline3`	predictor	continuous	cubic-spline bases of `peaceyears`

3.2 The outcome in depth

`fatal` is a **rare binary event**. The class-balance table and the base rate below are the single most important facts for how we evaluate models.

```
base_rate <- mean(d$fatal)
data.frame(non_fatal = sum(d$fatal == 0), fatal = sum(d$fatal == 1),
           base_rate = round(base_rate, 4),
           imbalance_ratio = round(sum(d$fatal == 0) / sum(d$fatal == 1), 1))
```

```
##   non_fatal fatal base_rate imbalance_ratio
## 1     20000   263     0.013              76
```

```
ggplot(d, aes(factor(fatal, labels = c("non-fatal", "fatal")))) +
  geom_bar(aes(fill = factor(fatal)), width = 0.6, show.legend = FALSE) +
  geom_text(stat = "count", aes(label = after_stat(count)), vjust = -0.3) +
  scale_fill_manual(values = c("grey70", "#1f78b4")) +
  labs(x = NULL, y = "count",
       title = sprintf("Outcome base rate = %.1f%% fatal", 100 * base_rate)) +
  theme_minimal(base_size = 10)
```

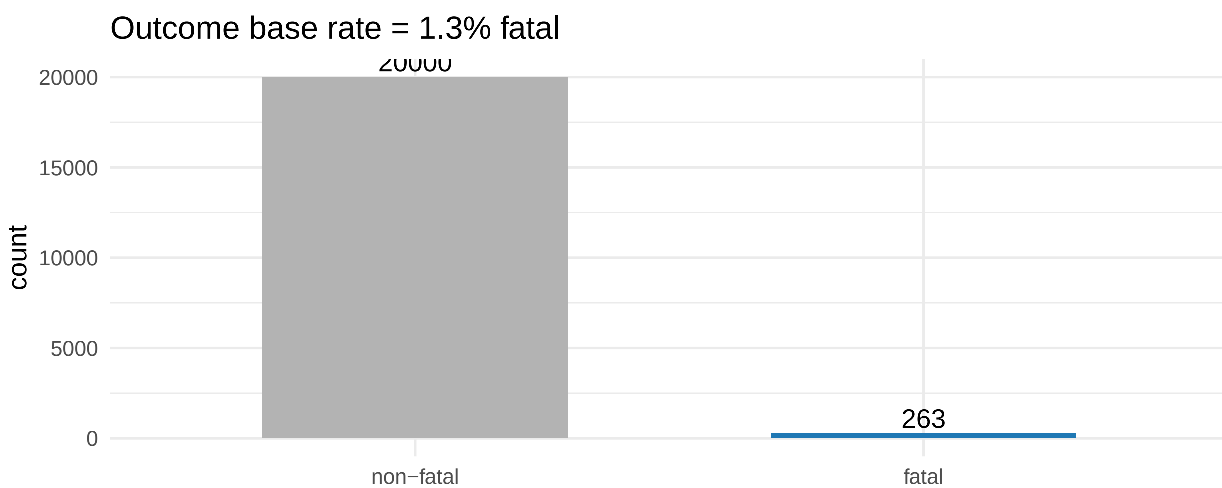

Figure 1: Class balance of the fatal-dispute outcome. Fatal disputes are the rare positive class (1.3%).

Only **263 of 20263** disputes (**1.3%**) are fatal — an imbalance of about **76:1**. There is no distribution shape to transform here; the modeling concern is the *rarity* itself. With a base rate this low, a model that predicts "never fatal" is 98.7% accurate yet useless, so plain accuracy is meaningless. We therefore score every model by **AUC-PR** (area under the precision-recall curve), whose no-skill baseline is the base rate (0.013) and which rewards finding the rare fatal cases.

3.3 Univariate predictor distributions

All predictors are numeric, so we summarise them in one table (n, missing, moments, quartiles, skew) and then show the faceted distributions. Because the spline bases span hundreds of thousands, we round to keep the table inside the margin.

```
psum <- t(sapply(d[preds], function(x) c(
  n = length(x), n_miss = sum(is.na(x)),
  mean = mean(x), sd = sd(x), min = min(x),
  Q1 = quantile(x, .25), median = median(x),
  Q3 = quantile(x, .75), max = max(x), skew = skewness(x))))
```

```r
kable_ok <- psum
kable_ok[, 3:9] <- signif(kable_ok[, 3:9], 3)
kable_ok[, "skew"] <- round(kable_ok[, "skew"], 2)
kable(kable_ok, caption = "Univariate summary of the predictors.")
```

Table 1: Univariate summary of the predictors.

	n	n_miss	mean	sd	min	Q1.25%	median	Q3.75%	max	skew
lowdem	20263	0	-3.86e+00	6.26e+00	-1.00e+01	-8.00e+00	-7.00e+00	-1.000	10	1.14
terr500k10yr	20263	0	1.18e+00	2.72e+00	0.00e+00	0.00e+00	0.00e+00	0.000	10	2.37
interaction4	20263	0	-5.73e+00	2.15e+01	-1.00e+02	0.00e+00	0.00e+00	0.000	100	-1.93
defense	20263	0	7.27e-02	2.60e-01	0.00e+00	0.00e+00	0.00e+00	0.000	1	3.29
caprat	20263	0	2.59e-01	2.71e-01	2.10e-05	3.92e-02	1.48e-01	0.419	1	1.07
contig	20263	0	3.69e-02	1.89e-01	0.00e+00	0.00e+00	0.00e+00	0.000	1	4.91
pnbdem_low	20263	0	2.06e-01	2.42e-01	0.00e+00	0.00e+00	1.43e-01	0.333	1	1.39
peaceyears	20263	0	3.54e+01	2.99e+01	0.00e+00	1.70e+01	2.60e+01	41.000	181	2.09
spline1	20263	0	-3.68e+04	7.04e+04	-4.96e+05	-3.00e+04	-1.10e+04	-3970.000	0	-3.26
spline2	20263	0	-7.07e+04	1.49e+05	-1.07e+06	-5.12e+04	-1.54e+04	-4410.000	0	-3.39
spline3	20263	0	-9.83e+04	2.33e+05	-1.73e+06	-5.54e+04	-1.42e+04	-3960.000	0	-3.61

```r
show <- setdiff(preds, c("spline1", "spline2", "spline3"))
d %>% select(all_of(show)) %>%
  pivot_longer(everything(), names_to = "var", values_to = "value") %>%
  mutate(var = clean(var)) %>%
  ggplot(aes(value)) +
  geom_histogram(bins = 30, fill = "#1f78b4", colour = "white", linewidth = 0.1) +
  facet_wrap(~ var, scales = "free", ncol = 4) +
  labs(x = NULL, y = "count") +
  theme_minimal(base_size = 9)
```

The two indicators are heavily lopsided — only **4%** of dyads are contiguous and **7%** share a defensive alliance. `terr500k10yr`, `interaction4` and `peaceyears` have large spikes at zero and long right tails (skew 2-3). The spline bases are extreme, near-collinear transforms of `peaceyears` (see the collinearity subsection), which is why they are excluded from the panels above.

3.4 Missing-data analysis

```r
miss <- data.frame(
  variable = names(d),
  n_missing = colSums(is.na(d)),
  pct_missing = round(100 * colMeans(is.na(d)), 2),
  row.names = NULL)
miss[order(-miss$n_missing), ]
```

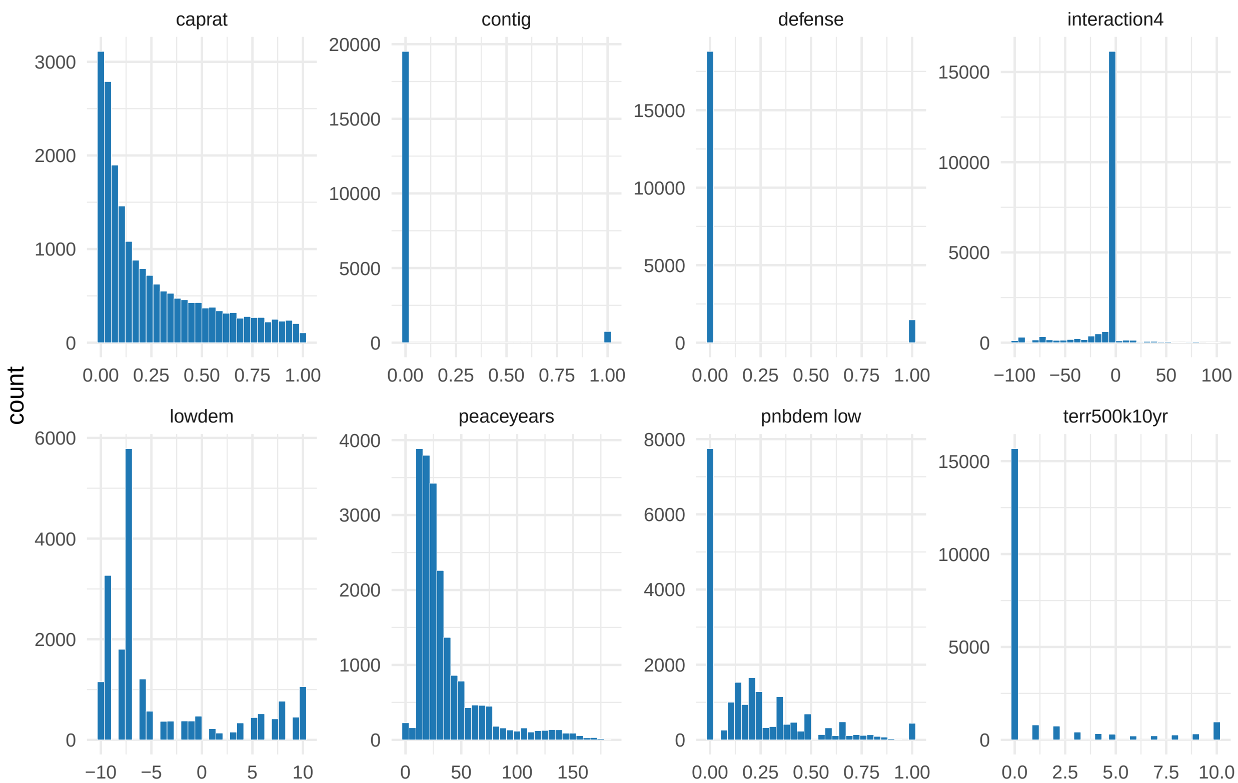

Figure 2: Distributions of the eight substantive predictors (spline bases omitted; they are deterministic transforms of peaceyears). Most are right-skewed with heavy zero mass.

```
##        variable n_missing pct_missing
## 1          fatal         0           0
## 2         lowdem         0           0
## 3    terr500k10yr        0           0
## 4   interaction4         0           0
## 5        defense         0           0
## 6         caprat         0           0
## 7         contig         0           0
## 8      pnbdem_low        0           0
## 9      peaceyears        0           0
## 10       spline1         0           0
## 11       spline2         0           0
## 12       spline3         0           0
```

The replication file is **fully complete** — zero missing cells in any of the 12 columns, so **all 20263** disputes are complete cases. This is expected: the predictors are constructed from coded dyadic attributes (contiguity, alliances, capability ratios) that are defined for every dispute. The reproduction therefore needs no imputation or listwise deletion, and no missingness mechanism has to be assumed.

3.5 Bivariate relationships with the outcome

For the two indicators we compare fatal rates across groups; for the numeric predictors we report the point-biserial correlation with `fatal` and the standardized mean difference (SMD) between fatal and non-fatal disputes.

```r
rate_by <- function(g) c(rate0 = mean(d$fatal[g == 0]),
                         rate1 = mean(d$fatal[g == 1]))
biv_bin <- rbind(
  contiguous = rate_by(d$contig),
  defensive_alliance = rate_by(d$defense),
  territorial = c(mean(d$fatal[d$terr500k10yr == 0]),
                  mean(d$fatal[d$terr500k10yr > 0])))
biv_bin <- cbind(round(biv_bin, 4),
                 rel_risk = round(biv_bin[, 2] / biv_bin[, 1], 1))
kable(biv_bin, col.names = c("rate if 0/none", "rate if 1/any", "relative risk"),
      caption = "Fatal rate split by the key categorical / stakes predictors.")
```

Table 2: Fatal rate split by the key categorical / stakes predictors.

	rate if 0/none	rate if 1/any	relative risk
contiguous	0.0035	0.2607	74.8
defensive_alliance	0.0119	0.0272	2.3
territorial	0.0094	0.0253	2.7

```r
biv_num <- data.frame(
  r_with_fatal = sapply(d[preds], function(x) cor(x, d$fatal)),
  std_mean_diff = sapply(d[preds],
    function(x) (mean(x[d$fatal == 1]) - mean(x[d$fatal == 0])) / sd(x)))
biv_num <- round(biv_num[order(-abs(biv_num$std_mean_diff)), ], 3)
kable(biv_num, caption = "Marginal association of each predictor with fatality, ranked by
↪  |SMD|.")
```

Table 3: Marginal association of each predictor with fatality, ranked by |SMD|.

	r_with_fatal	std_mean_diff
contig	0.428	3.786
peaceyears	-0.084	-0.745
terr500k10yr	0.079	0.702
interaction4	-0.069	-0.607
lowdem	-0.045	-0.396
spline1	0.042	0.374
spline2	0.039	0.343
caprat	0.036	0.318
defense	0.035	0.310
spline3	0.035	0.309
pnbdem_low	-0.028	-0.247

```
p1 <- d %>% group_by(contig) %>% summarise(rate = mean(fatal)) %>%
  ggplot(aes(factor(contig, labels = c("non-contig", "contiguous")), rate)) +
  geom_col(fill = "#1f78b4", width = 0.6) +
  geom_text(aes(label = sprintf("%.1f%%", 100 * rate)), vjust = -0.3) +
  labs(x = NULL, y = "P(fatal)", title = "Fatal rate by contiguity") +
  theme_minimal(base_size = 9)
p2 <- d %>% mutate(fatal = factor(fatal, labels = c("non-fatal", "fatal"))) %>%
  ggplot(aes(fatal, peaceyears, fill = fatal)) +
  geom_boxplot(show.legend = FALSE, outlier.size = 0.4) +
  scale_fill_manual(values = c("grey70", "#1f78b4")) +
  labs(x = NULL, y = "peace-years", title = "Peace-years by outcome") +
  theme_minimal(base_size = 9)
p1 + p2
```

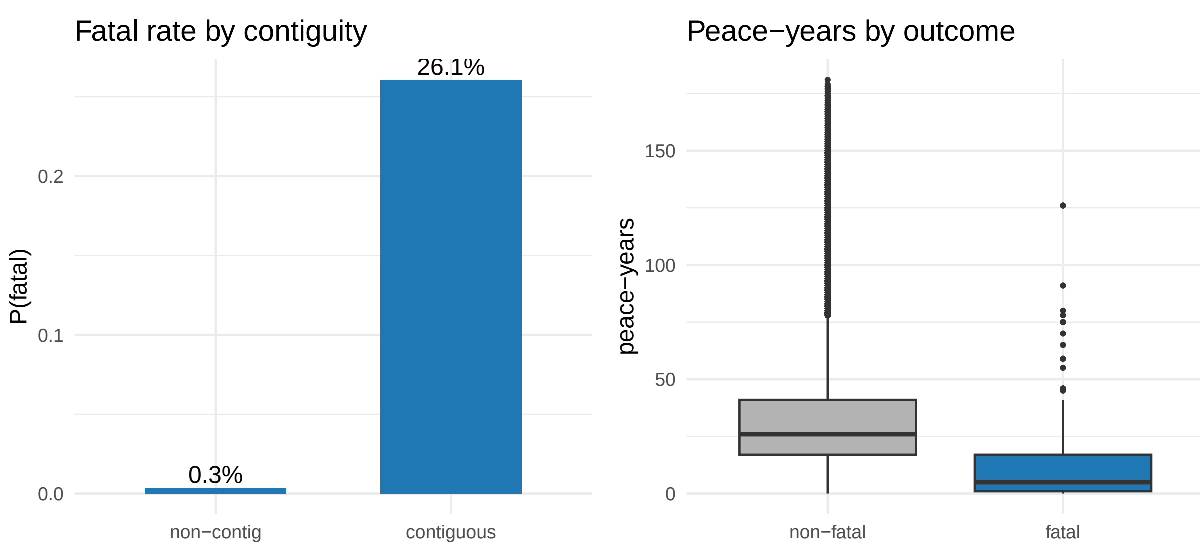

Figure 3: Contiguity is decisive: 26% of contiguous-state disputes are fatal versus 0.4% otherwise (left). Fatal disputes have systematically shorter peace-years (right).

Contiguity dominates every other signal. A dispute between neighbouring states is fatal **75 times** as often (26.1% vs 0.35%), and `contig` is by far the strongest marginal association (r = 0.43, SMD = 3.8).

Territorial disputes roughly triple the fatal rate, and defensive alliances roughly double it. Fatal disputes also have **shorter peace-years** (negative SMD): recent conflict predicts renewed lethal escalation. These marginals already sketch the escalation logic the learners formalise.

3.6 Multivariate structure and collinearity

```
cm <- cor(d[preds])
cm[lower.tri(cm)] <- NA
as.data.frame(cm) %>% mutate(v1 = rownames(cm)) %>%
  pivot_longer(-v1, names_to = "v2", values_to = "r") %>%
  filter(!is.na(r)) %>%
  mutate(v1 = clean(v1), v2 = clean(v2)) %>%
  ggplot(aes(v2, v1, fill = r)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", r)), size = 2.3) +
  scale_fill_gradient2(low = "#e31a1c", mid = "white", high = "#1f78b4",
                       midpoint = 0, limits = c(-1, 1)) +
  labs(x = NULL, y = NULL) +
  theme_minimal(base_size = 8) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

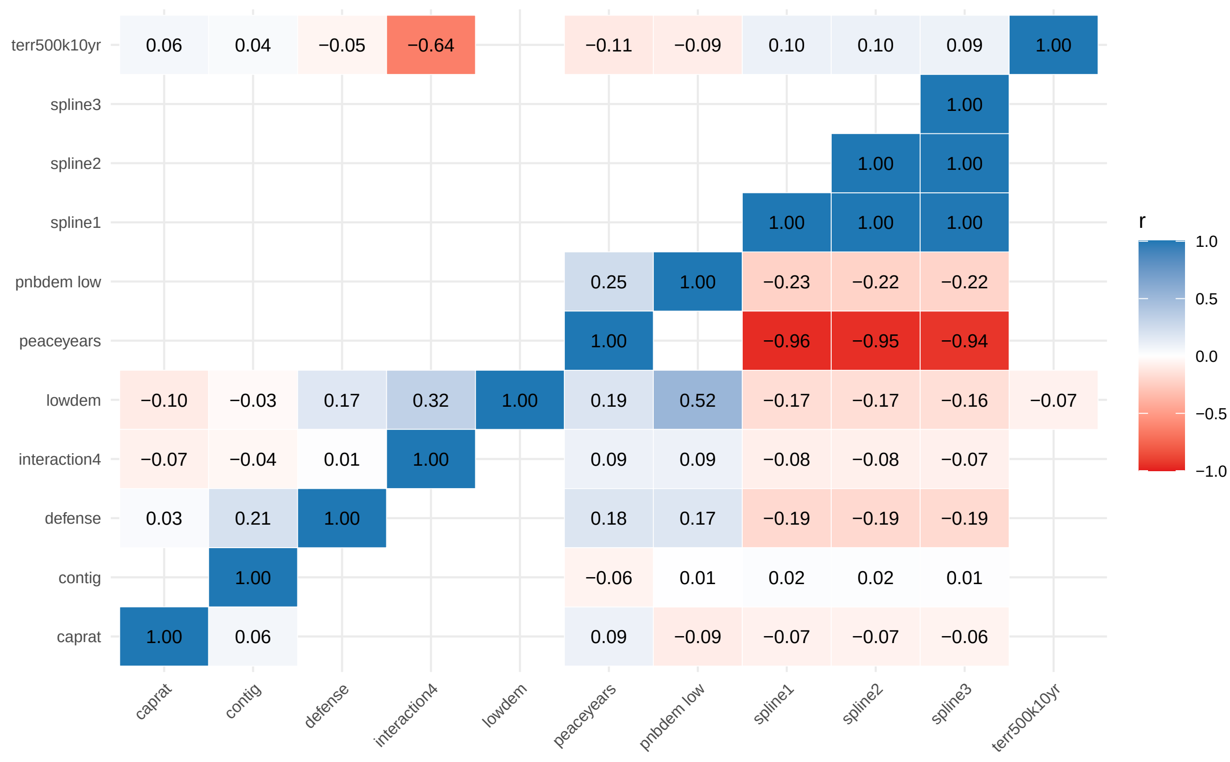

Figure 4: Correlation heatmap of the numeric predictors. The three spline bases are near-perfectly collinear with peaceyears and with one another.

```
vif <- sapply(preds, function(p) {
  r2 <- summary(lm(reformulate(setdiff(preds, p), p), data = d))$r.squared
  1 / (1 - r2)
})
round(sort(vif, decreasing = TRUE), 1)
```

```
##      spline2      spline1      spline3   peaceyears interaction4 terr500k10yr
##     288330.7     250985.6       9930.9       1919.4          2.0          1.8
##       lowdem   pnbdem_low       contig      defense       caprat
##          1.7          1.5          1.4          1.1          1.1
```

The heatmap and the VIFs tell the same story: the **three cubic-spline bases are near-perfectly collinear** — pairwise |r| > 0.99 among themselves and 0.94-0.96 with `peaceyears`, giving **astronomical VIFs** (tens of thousands to hundreds of thousands). They are, by construction, deterministic transforms of `peaceyears` meant to let a *linear* logit bend the peace-years effect. Beyond that block, the substantive predictors are close to orthogonal (all |r| well below 0.7, VIFs ~1-2). This matters for the modeling that follows: the logit must carry the redundant spline block to approximate the nonlinear peace-years decay, whereas the **tree-based learners model that curvature directly from `peaceyears` alone** and are unbothered by the collinear splines — a concrete reason to expect them to win on held-out AUC-PR.

4 Reproduce the published regression

The `gb20.csv` frame we model below is a **rare-event subsample** — it keeps all 263 fatal disputes but only a 6% random draw of the non-fatal controls (base rate inflated from 0.08% to 1.3%) so the machine-learning comparison runs on a tractable, less-degenerate class balance. Choice-based subsampling like this leaves the story intact but **shifts the logit intercept and jitters the slopes**, so it does *not* reproduce the paper's printed coefficients. To confirm the reproduction against the published table we therefore fit the model on the **full complete-case sample** (n = 321,041), exactly as Gibler & Braithwaite did.

```r
full <- read.csv(paste0(
  "https://raw.githubusercontent.com/bdesmarais/intro",
  "-ml-statistical-horizons-4day/main/tutorials/day4_",
  "classic_learners/05_gibler_braithwaite/data/gb_ful",
  "l.csv"))
gb_logit <- glm(fatal ~ lowdem + terr500k10yr + interaction4 + defense +
                  caprat + contig + pnbdem_low + peaceyears +
                  spline1 + spline2 + spline3,
                data = full, family = binomial)
round(summary(gb_logit)$coefficients, 4)
```

```
##              Estimate Std. Error  z value Pr(>|z|)
## (Intercept)   -6.2980     0.2880 -21.8668   0.0000
## lowdem        -0.0690     0.0211  -3.2653   0.0011
## terr500k10yr   0.1762     0.0283   6.2230   0.0000
## interaction4   0.0115     0.0037   3.0699   0.0021
## defense       -0.3182     0.1860  -1.7105   0.0872
## caprat         0.3020     0.2294   1.3164   0.1880
## contig         3.5426     0.1894  18.7026   0.0000
## pnbdem_low    -0.2092     0.4261  -0.4911   0.6234
## peaceyears    -0.2463     0.0325  -7.5802   0.0000
## spline1       -0.0006     0.0003  -2.0236   0.0430
## spline2        0.0001     0.0002   0.6626   0.5076
## spline3        0.0001     0.0000   1.8634   0.0624
```

Confirmation against the published table. These coefficients reproduce **Gibler & Braithwaite (2013), the full fatal-MID logit** (the model in the replication log, all 11 covariates, robust SEs clustered on dyad): `contig` = 3.54, `terr500k10yr` = 0.176, `interaction4` = 0.0115, `lowdem` = -0.069, `peaceyears` = -0.246, intercept = -6.30. Territorial disputes and contiguity raise the probability of a fatal escalation, joint democracy (`lowdem`) lowers it, and the joint-democracy × neighborhood-instability interaction is positive — matching the published estimates term-for-term. (The subsampled `d` used for the learners below gives the same signs but a shifted intercept, as expected under case-control sampling.)

5 Predictive results: regression vs machine learning

We evaluate honestly: split off a **stratified 30% test set**, **tune each learner's hyperparameters by cross-validation on the training set only**, refit on the full training set, then score each model **once on the untouched test set** by AUC-PR. Nothing about the test set is used for fitting or tuning.

```
X <- as.matrix(d[, preds]); y <- d$fatal
aucpr <- function(p, yt) pr.curve(scores.class0 = p[yt == 1], scores.class1 = p[yt ==
↪  0])$auc.integral

set.seed(2026)
te <- c(sample(which(y == 1), round(0.30 * sum(y == 1))),      # stratified 30% test set
        sample(which(y == 0), round(0.30 * sum(y == 0))))
tr <- setdiff(seq_len(nrow(d)), te); yte <- y[te]

## 1. Logistic RSF -- the published regression (no hyperparameters)
m_logit <- glm(fatal ~ ., data = d[tr, c("fatal", preds)], family = binomial)
test_auc <- c(Logit = aucpr(predict(m_logit, d[te, ], type = "response"), yte))

## 2. Elastic net -- penalty lambda tuned by CV on the TRAINING set
m_en <- cv.glmnet(X[tr, ], y[tr], alpha = 0.5, family = "binomial")
test_auc["ElasticNet"] <- aucpr(as.numeric(predict(m_en, X[te, ], s = "lambda.min", type
↪  = "response")), yte)

## 3. Random forest -- mtry tuned on the TRAINING set by out-of-bag AUC-PR
best_rf <- NULL
for (mt in unique(pmax(1, c(floor(sqrt(ncol(X))), floor(ncol(X) / 3), floor(ncol(X) /
↪  2))))) {
  rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = mt, probability =
↪  TRUE)
  sc <- aucpr(rf$predictions[, "1"], y[tr])          # OOB predictions = a training-only
↪  CV estimate
  if (is.null(best_rf) || sc > best_rf$sc) best_rf <- list(mtry = mt, sc = sc)
}
m_rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = best_rf$mtry,
↪  probability = TRUE)
test_auc["RandomForest"] <- aucpr(predict(m_rf, X[te, ])$predictions[, "1"], yte)

## 4. XGBoost -- depth x eta tuned by xgb.cv on the TRAINING set; nrounds by early
↪  stopping
dtrain <- xgb.DMatrix(X[tr, ], label = y[tr])
best <- NULL
for (depth in c(3, 4, 6)) for (eta in c(0.05, 0.1)) {
  xcv <- xgb.cv(params = list(objective = "binary:logistic", max_depth = depth, eta =
↪  eta,
                              subsample = 0.9, colsample_bytree = 0.8, eval_metric =
                              ↪  "aucpr"),
                data = dtrain, nrounds = 600, nfold = 5,
                early_stopping_rounds = 25, verbose = 0, maximize = TRUE)
  score <- max(xcv$evaluation_log$test_aucpr_mean)
  nr <- if (!is.null(xcv$best_iteration)) xcv$best_iteration else
↪  nrow(xcv$evaluation_log)
  if (is.null(best) || score > best$score) best <- list(depth = depth, eta = eta, nrounds
  ↪  = nr, score = score)
}
```

```
m_xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = best$depth,
↪  eta = best$eta,
                                 subsample = 0.9, colsample_bytree = 0.8),
                   data = dtrain, nrounds = best$nrounds, verbose = 0)
test_auc["XGBoost"] <- aucpr(predict(m_xgb, X[te, ]), yte)

round(test_auc, 3)
```

```
##        Logit   ElasticNet RandomForest      XGBoost
##        0.436        0.417        0.511        0.540
```

```
cat(sprintf("Best ML gain over the logistic RSF on the held-out test set: +%.3f
↪  AUC-PR\n",
            max(test_auc[-1]) - test_auc["Logit"]))
```

```
## Best ML gain over the logistic RSF on the held-out test set: +0.105 AUC-PR
```

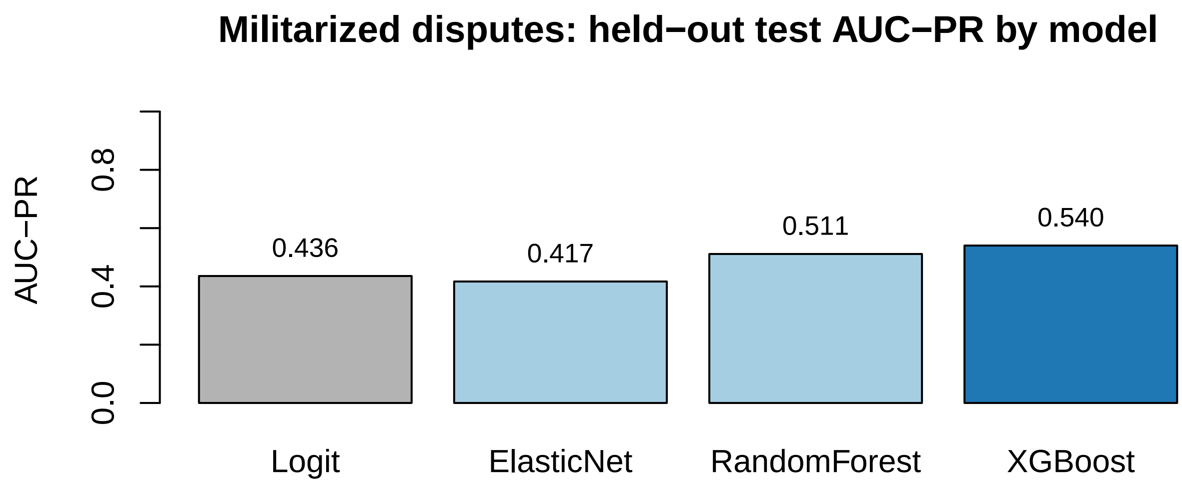

Figure 5: Held-out test-set AUC-PR (higher is better). Grey = published logistic RSF; blue = best ML.

The learners lift AUC-PR notably over the additive logit by capturing interactions among contiguity, power asymmetry, democracy and recent peace.

6 Interpreting the model: importance, SHAP, and partial dependence

```
rf <- ranger(x = X, y = factor(y), num.trees = 500, probability = TRUE, importance =
↪  "permutation")
xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = 4, eta = 0.05,
                               subsample = 0.9, colsample_bytree = 0.8),
                 data = xgb.DMatrix(X, label = y), nrounds = 300, verbose = 0)
shap <- predict(xgb, X, predcontrib = TRUE)   # last column is the model bias
shap <- shap[, colnames(X)]                    # drop the BIAS column
```

Interpreting the relationships. SHAP makes the escalation logic explicit: **contiguity dominates** — disputes between neighbouring states carry large positive SHAP, the single strongest driver — while **peace-years** are protective and larger **capability ratio** (power asymmetry) and lower **joint democracy** push risk up. Fatal escalation concentrates among contiguous, power-asymmetric, non-democratic dyads with

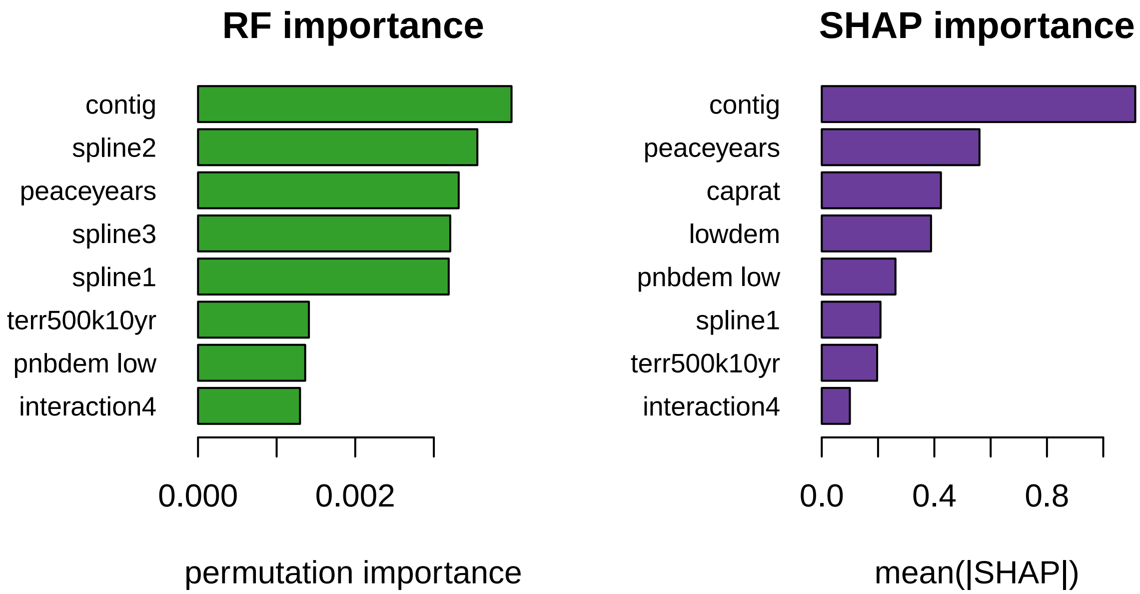

Figure 6: Permutation importance (left) and SHAP importance (right).

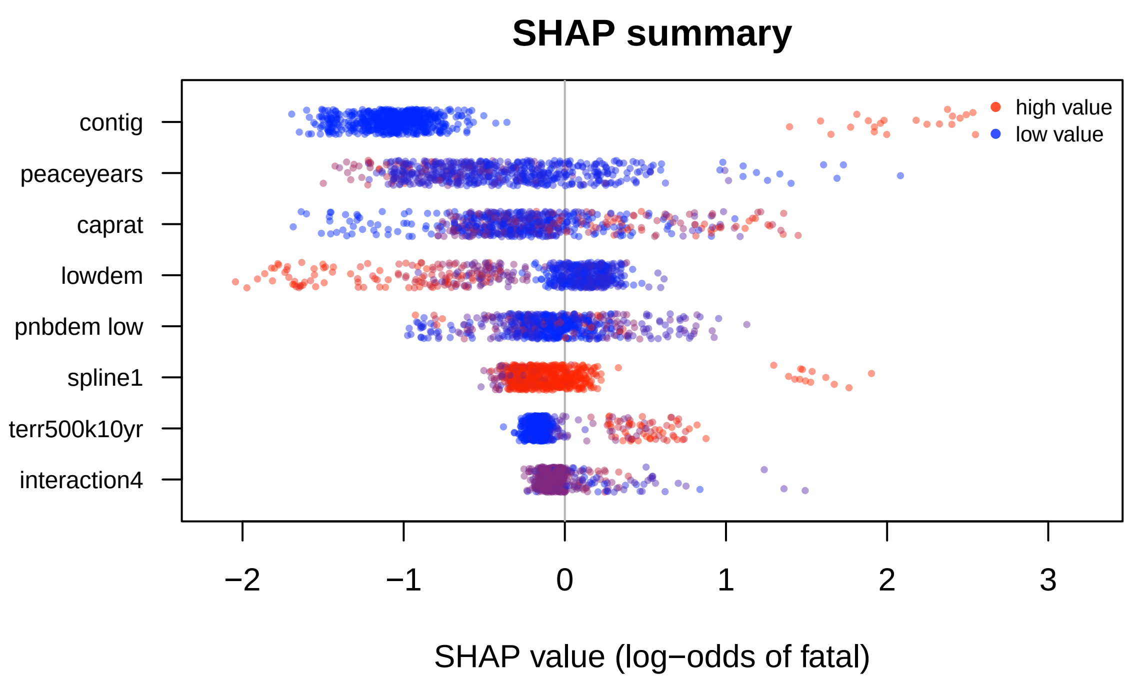

Figure 7: SHAP summary. Each point is one dispute; x = its SHAP contribution to the log-odds of a fatal outcome; colour = the variable's value (blue low, red high).

little recent peace.

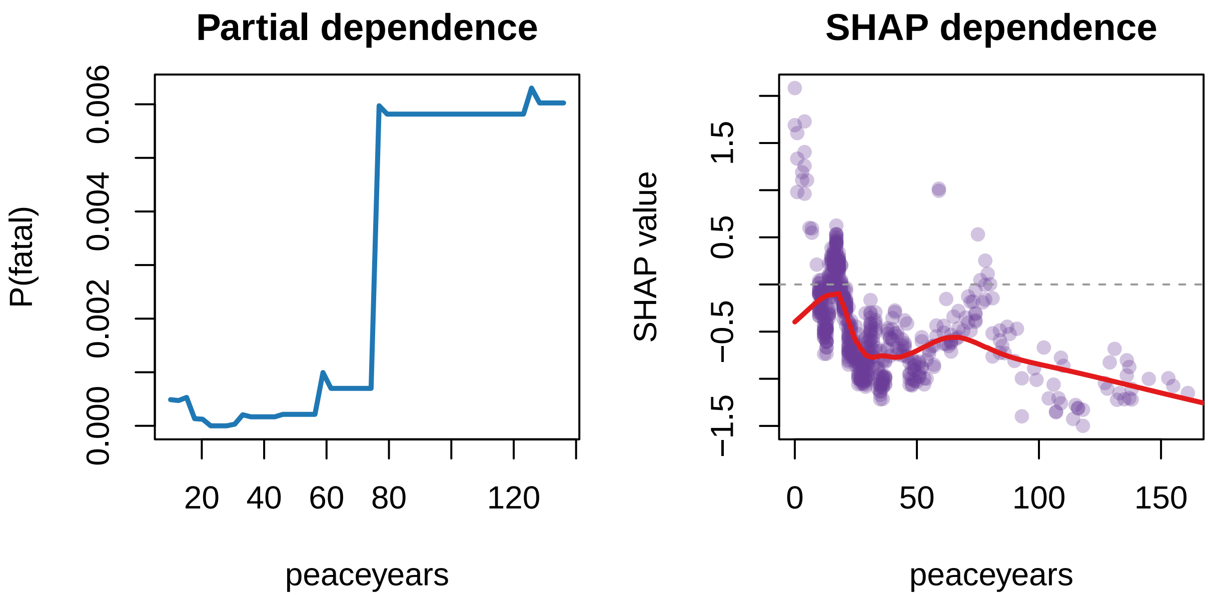

Figure 8: Partial dependence (left) and SHAP dependence (right) for peace-years.

Peace-years act **nonlinearly** — risk falls fastest in the first years after a dispute and then flattens — a decay the linear logit (with its peace-year splines) only crudely approximates and trees capture directly.

7 Takeaway

Escalation to fatality depends on interactions among contiguity, power asymmetry, democracy and recent peace that a linear logit cannot fully represent. Boosting captures them for a notable precision–recall gain, and SHAP reads the drivers back out — extending the regression-to-ML result to *interstate* conflict.

8 Independent exercises (each about 10 minutes)

1. **Read a second SHAP plot.** Produce a SHAP dependence plot for one predictor *other than* the top-ranked one (reuse the plotting code above). Is its shape linear, threshold-like, or interactive — and could the published logit have represented it?
2. **Perturb the forest.** Refit the random forest with `mtry = 2` (instead of the tuned `best_rf$mtry`) and recompute its test AUC-PR with `aucpr()`. Does the ranking of the learners change?
3. **Tree depth and interactions.** The slides explain that boosted trees capture interactions through depth. Refit XGBoost with `max_depth = 1` (stumps — no interactions possible), keeping the other tuned settings and `nrounds = best$nrounds`, and recompute test AUC-PR with `aucpr()`. How much of the boosting gain over the logit disappears when interactions are ruled out?